HEMA RAJPUT

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PROFESSIONAL SUMMARY

Artificial Intelligence & Data Science undergraduate specializing in **Machine Learning, Deep Learning, Generative AI, Retrieval-Augmented Generation (RAG),** Skilled in building end-to-end, production-style AI systems using **Python, React.js, FastAPI, PostgreSQL**, and modern AI frameworks. Recognized nationally through the **IEEE Space B.Tech Initiative** and the **Bharatiya Antariksh Hackathon** for innovation and technical excellence.

TECHNICAL SKILLS

Programming Languages: Python
Web Technologies: HTML, CSS, React.js
AI & Machine Learning: Machine Learning (Regression, Classification, Clustering), Deep Learning (Neural Networks, CNNs, RNNs, Transformers), Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Agentic AI
Generative AI & Frameworks: LangChain, LangGraph, MCP (Model Context Protocol)
Data Science & Analytics: NumPy, Pandas, Matplotlib, Seaborn, Data Cleaning, Data Analysis, Data Visualization
Backend & APIs: FastAPI, REST APIs, JWT Authentication
Databases: PostgreSQL, MongoDB
DevOps & Deployment: Docker, CI/CD

PROFESSIONAL EXPERIENCE

CyberPeace Foundation — Research Wing / *AI Research Contributor*

- Developed an **AI-powered toll monitoring system** to automate vehicle detection, license plate recognition, transaction verification, and fraud monitoring.
- Built an end-to-end **ANPR pipeline** using **YOLOv11, EasyOCR, and OpenCV** for real-time vehicle and license plate detection, number-plate extraction, and automated verification.
- Developed **FastAPI REST APIs** and integrated a **35-table PostgreSQL database** for managing vehicles, FASTag records, transactions, fraud cases, alerts, and system data.
- Built a **React-based operational dashboard** for live monitoring, transaction tracking, fraud alerts, analytics, and system health visualization.
- Implemented **AI-driven and rule-based validation** to flag suspicious activity, including plate mismatches, blacklisted vehicles, and abnormal transactions.

Programmates Pvt. Ltd. / *AI & Data Science Intern*

- Worked on **machine learning workflows**, data analysis, and software development tasks.
- Assisted in implementing **data-driven solutions** and supporting AI-related development activities.
- Applied **analytical and problem-solving skills** to assigned technical tasks.
- Collaborated on software development activities with a proactive, solution-oriented approach.

TECHNICAL PROJECTS

AI-Based Intelligent Toll Detection & Fraud Prevention System / *Python, YOLOv11, EasyOCR, OpenCV, FastAPI, React, PostgreSQL*

- Built a full-stack **AI-powered toll monitoring system** using **YOLOv11 and EasyOCR** for real-time vehicle and license plate detection.
- Developed a **React dashboard** and **FastAPI backend** for live monitoring, transaction management, fraud detection, alerts, and analytics.
- Integrated a **PostgreSQL database with 35 interconnected tables** for centralized system data management.

G.W.E.N. — Personalized Voice, Vision, Gesture & Automation Assistant / *Python, Ollama, Whisper, OpenCV, LangChain, LangGraph*

- Developed a **multimodal AI assistant** integrating voice interaction, computer vision, gesture recognition, contextual memory, and desktop automation.
- Integrated **Whisper, Ollama LLMs, Piper TTS, and OpenCV** to enable speech understanding, intelligent responses, voice output, face recognition, and visual interaction.
- Implemented **LangChain/LangGraph-based task routing** with **semantic memory, reminders, face authentication, system control, and personalized task execution**.

ACHIEVEMENTS

- **Bharatiya Antariksh Hackathon 2025 (ISRO & iHUB)** — Selected among top participants nationally for innovative space technology solutions.
- **IEEE Space B.Tech Initiative** — Selected for IEEE's prestigious space-track undergraduate program.
- **Hacksagon Hackathon** — Finalist.
- **ET AI Hackathon** — Finalist. Advanced to final rounds in two national AI hackathons, evaluated by industry experts.